LABORATORY REPORT

Application Development Lab
(CS33002)

B.Tech Program in CSE

Submitted By

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(Deemed to be University)
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Experiment Number	1

Experiment Title	Introduction to Machine Learning
Date of Experiment	07/01/2025
Date of Submission	14/01/2025

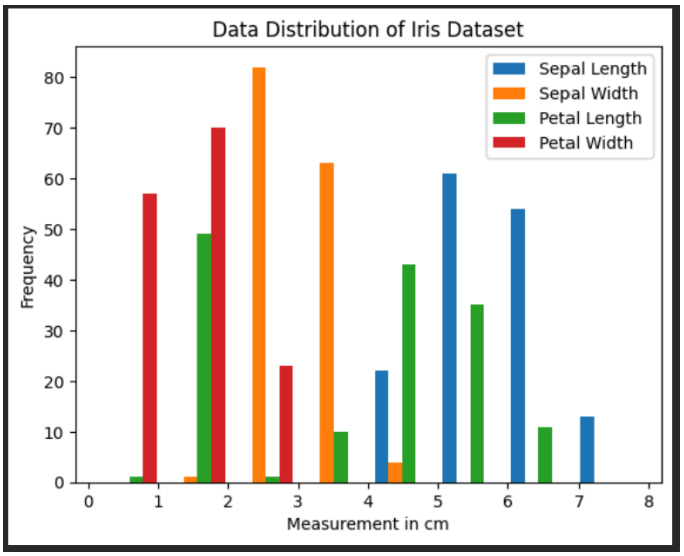
1. Objective: Understand the basics of machine learning and Python/R

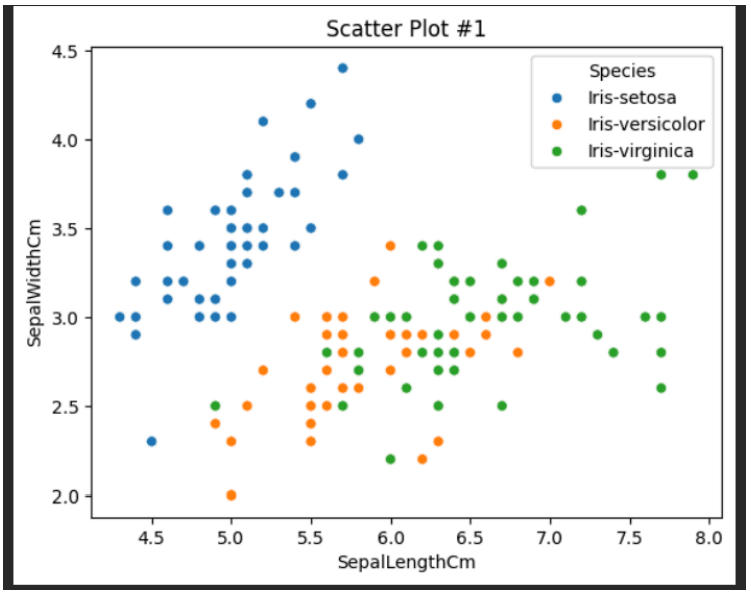
2. Code:-

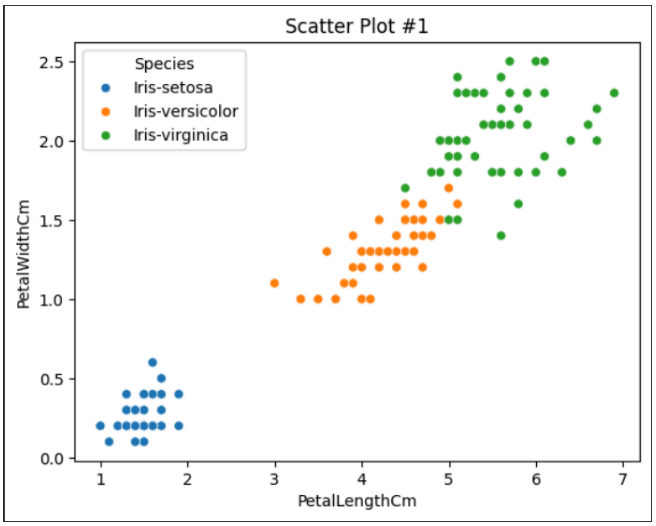
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
import pandas as pd
iris = load_iris()
df = pd.DataFrame(
   iris.data,
 columns=['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']
df['Species'] = iris.target
df['Species'] = df['Species'].map({
    0: 'Iris-setosa',
    1: 'Iris-versicolor',
    2: 'Iris-virginica'
})
df.info()
df.describe()
df.isnull().sum()
df.insert(0, 'Id', range(1, len(df) + 1))
df.set_index('Id', inplace=True)
plt.hist(
    df[['SepalLengthCm','SepalWidthCm','PetalLengthCm','PetalWidthCm']],
    bins=8
)
plt.xlabel("Measurement in cm")
plt.ylabel("Frequency")
plt.title("Data Distribution of Iris Dataset")
plt.legend(['Sepal Length', 'Sepal Width', 'Petal Length', 'Petal Width'])
plt.show()
sns.scatterplot(
    x='SepalLengthCm',
    y='SepalWidthCm',
    hue='Species',
    data=df
)
plt.title("Scatter Plot #1")
plt.show()
sns.scatterplot(
```

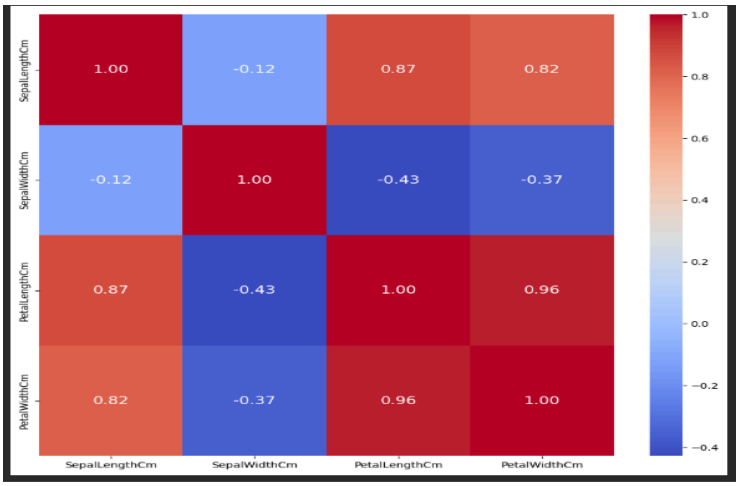
```
 x='PetalLengthCm',
    y='PetalWidthCm',
    hue='Species',
    data=df
)
plt.title("Scatter Plot #1")
plt.show()
ndf = df.select_dtypes(include='number')
corr = ndf.corr()
corr
plt.figure(figsize=(11,10))
sns.heatmap(
   corr,
   annot=True,
   annot_kws={"size": 14},
cmap='coolwar
fmt='.2f')plt.show()
```

3. Results/Output:- Entire Screen Shot including Date & Time









4. Remarks:-

Signature of the Student _____ (Name of the Student)	Signature of the Lab Coordinator _____ __ (Name of the Coordinator)